

SENG 513 - PoetryVerse Database Schema and Seeding Script - PG 2

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Schema File

The Prisma database schema is located at packages/database/prisma/schema.prisma.

Seeding Script

The seed script is located at packages/database/prisma/seed.js.

Seed data is stored in packages/database/prisma/seeding-data/.

Prerequisites

Install Git from <https://git-scm.com/install/>.

Install and run Docker from <https://docs.docker.com/get-started/get-docker/>.

We recommend using Visual Studio Code for development, which can be installed from <https://code.visualstudio.com/download>.

Step 1: Clone the repository

Clone the PoetryVerse project repository with SSH with this command:

- `git clone git@csgit.ualgary.ca:rohan.kapila/seng513-202601-pg-2.git`

Navigate into the project folder with this command:

- `cd seng513-202601-pg-2`

Switch to the commit tagged with 'schema-milestone' with this command:

- `git switch --detach schema-milestone`

Step 2: Create the .env files

Create the .env and .env.production files by copying their respective .env.example and .env.production.example files with these commands:

- `cp .env.example .env`
- `cp .env.production.example .env.production`
- `cp apps/api/.env.example apps/api/.env`
- `cp apps/api/.env.production.example apps/api/.env.production`
- `cp apps/web/.env.example apps/web/.env`
- `cp apps/web/.env.production.example apps/web/.env.production`
- `cp packages/database/.env.example packages/database/.env`

Step 3: Populate the .env files

Note to the TAs: Please populate the .env files with the .env file examples we have submitted.

Otherwise, if the .env file examples were not provided, you would populate the .env files created in the previous step by following their commented instructions. You will need to contact a PoetryVerse team member to obtain some of the secret environment variable values.

Schema and seeding instructions with the Docker PostgreSQL instance

[Compose] Step 4.1: Run PoetryVerse with Docker Compose

Start the *development* Compose environment with this command:

- `docker compose up --watch`

Alternatively, start the *production* Compose environment with this command:

- `docker compose --env-file .env.production -f docker-compose.yml -f docker-compose.prod.yml up`

Note: When spinning up either environment, the “migrate” service defined in the base `docker-compose.yml` gets created, applies the migrations and the schema on the database, and exits.

Note: The Compose environments must be stopped and cleaned up before switching between *development* and *production*:

- Stop and clean up a *development* environment with this command:
 - `docker compose down`
- Stop and clean up a *production* environment with this command:
 - `docker compose -f docker-compose.yml -f docker-compose.prod.yml down`

[Compose] Step 5.1: Seed the containerized PostgreSQL database

Seed the database in the *development* Compose environment with this command:

- `docker compose run --rm migrate npm run db:seed --workspace=packages/database`

Alternatively, seed the database in the *production* Compose environment with this command:

- `docker compose --env-file .env.production -f docker-compose.yml -f docker-compose.prod.yml run --rm migrate npm run db:seed --workspace=packages/database`

Note: Running the seed command populates seed data by starting a temporary container based on the “migrate” service and overrides the service’s default command with the Prisma seeding command.

[Compose] Step 6.1: View the database contents

Connect to the containerized *development* PostgreSQL database with this command:

- `docker exec -it poetryverse-db psql -U poetryverse -d poetryverse`

Alternatively, connect to the containerized *production* PostgreSQL database, replacing *POSTGRES_USER* and *POSTGRES_DB* with their secret values in the root *.env.production* file, with this command:

- `docker exec -it poetryverse-db psql -U <POSTGRES_USER> -d <POSTGRES_DB>`

In the psql terminal, list the PoetryVerse database tables with this command:

- `\dt`

View a table's schema with this command:

- `\d "<table>"`
- For example:
 - `\d "Poem"`
- *Tip:* If necessary, use `q` to exit the viewer

View the seeded rows for a table with this command:

- `SELECT * FROM "<table>";`
- For example:
 - `SELECT * FROM "Poem";`
- *Tip:* If necessary, use `q` to exit the viewer
- *Tip:* For a cleaner view of the table rows, enable expanded display with this command:
 - `\pset x on`

Schema and seeding instructions with a local PostgreSQL database

Prerequisite: Install PostgreSQL and set up a local database:

- <https://www.postgresql.org/download/>

Important Note: Ensure the *DATABASE_URL* environment variable in *packages/database/.env* is uncommented and points to your local PostgreSQL database.

[Local] Step 4.2.1: Install dependencies and generate the Prisma client

Install the app's dependencies with this command:

- `npm install`

Generate the Prisma client with this command:

- `npm run db:generate --workspace=packages/database`

[Local] Step 4.2.2: Apply schema/migrations to the local database

There are several ways to apply the schema/migrations to your *local* database.

The easiest is with this command, which just applies the Prisma schema directly to your database (recommended for basic local setup):

- `npm run db:push --workspace=packages/database`

Alternatively, the schema can be applied by running the Prisma migrations with this command (recommended for CI/staging/production):

- `npm run db:deploy --workspace=packages/database`

[Local] Step 5.2: Seed the local database

Once the Prisma schema is applied, you can run the Prisma seeding script with this command:

- `npm run db:seed --workspace=packages/database`

[Local] Step 6.2: View the database contents

Connect to the *local* PostgreSQL database using your corresponding *username* and *db_name* with this command:

- `psql -U <username> -d <db_name>`

In the psql terminal, list the PoetryVerse database tables with this command:

- `\dt`

View a table's schema with this command:

- `\d "<table>"`
- For example:
 - `\d "Poem"`
- *Tip:* If necessary, use `q` to exit the viewer

View the seeded rows for a table with this command:

- `SELECT * FROM "<table>";`
- For example:
 - `SELECT * FROM "Poem";`
- *Tip:* If necessary, use `q` to exit the viewer

- *Tip:* For a cleaner view of the table rows, enable expanded display with this command:
 - `\pset x on`
- *Important Note:* The Account and Session tables are not seeded, as they store session and authentication data for active client connections